

Minimal atelectasis and poorly aerated lung on thoracic CT images of normal dogs acquired under sedation

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Previous presentation: Portions of the results of this paper were previously presented as an abstract at the American College of Veterinary Radiology Annual Meeting Baltimore, Maryland 2019

Equator network checklist disclosure: No reporting guidelines were used

Abstract

General anesthesia optimizes image quality of thoracic CT in dogs by providing patient immobilization and respiratory control. However, it also comes with disadvantages that may serve as a detractor to the use of CT in veterinary practice. With the use of multidetector helical CT combined with injectable sedation for restraint, parameters such as atelectasis and poorly aerated lung should not be appreciably different from what has been observed in dogs under general anesthesia. This prospective pilot study aimed to evaluate the mean lung attenuation and the amount of atelectasis and poorly aerated lung on thoracic CT images acquired under injectable sedation in normal dogs. The entire thorax of 10 client-owned dogs was scanned. Attenuation measurements were generated by three-dimensional reconstruction software. Mean $\pm$ SD lung attenuation was -707.0 ± 60.0 . Atelectasis was not identified on any of the scans. Hypoinflated lung, the percentage of lung parenchyma with attenuation greater than -500 and -250 Hounsfield units (HU), was $10.7 \pm 4.7\%$ and $2.4 \pm 1.2\%$ (mean $\pm$ SD), respectively. There was no significant change in these percentages over time. Compared to previously published data, thoracic CT images obtained under sedation had mean attenuation comparable to normal expiratory lung and a lower percentage of poorly aerated lung compared to that of anesthetized dogs. Using sedation to complete canine thoracic CT does not itself lead to alterations of lung attenuation and may confer less hypoinflation and atelectasis than general anesthesia.

KEYWORDS

grow-from-seed segmentation, HAA%, high attenuation area, hypoinflated lung, MDCT

1 | INTRODUCTION

Computed tomography (CT) is an established modality for diagnosing thoracic disease in small animal medicine.^{1–3} Computed tomography performed under general anesthesia has been recommended in order to maximize image quality but disadvantages include risks

of anesthetic complications^{4,5} and added costs. Other factors that may degrade CT image quality include patient movement, respiratory motion, and atelectasis, all of which could mimic or obscure pathology. General anesthesia is employed to control patient movement,^{6,7} and mechanical ventilation, breath-holding, or induced apnea techniques must be used to halt respiratory motion.^{8,9} Atelectasis is ameliorated or prevented by recruitment maneuvers and positive pressure ventilation^{10–12}; although, these techniques cannot completely eliminate atelectasis commonly present in thoracic CT images.

General anesthesia often creates the conditions under which atelectasis develops. Studies in humans and dogs have shown that atelectasis is present on CT images immediately after induction, particularly

Abbreviations: ACVECC, American College of Veterinary Emergency & Critical Care; BCS, body condition score; FiO₂, fraction of inspired oxygen; HAA%–250, percentage of high attenuating lung area greater than -250 Hounsfield units; HAA%–500, percentage of high attenuating lung area greater than -500 Hounsfield units; HU, Hounsfield units; MDCT, multi-detector helical CT; PEEP, positive end-expiratory pressure; PP, positive pressure ventilation; ROI, region of interest; SD, standard deviation

when neuromuscular blockade is used.^{13–15} Passive compression of pulmonary tissues and the accompanying increased intrathoracic pressure due to respiratory muscle relaxation leads to compressive atelectasis. Atelectasis also occurs when oxygen diffuses from the alveolar space into the pulmonary capillaries faster than ventilation replaces the volume of alveolar gas.^{15,16} Diffusion atelectasis is reduced in both humans and veterinary patients maintained on a fraction of inspired oxygen (FiO₂) of 0.40 compared to those breathing FiO₂ of 1.0.^{16–18} Computed tomography scans performed on human children under injectable sedation show little to no atelectasis as compared to anesthetized subjects.^{19,20} Multi-detector helical CT (MDCT) machines of 32 slices or higher have faster scan times than prior generations.⁹ A complete thoracic scan can be completed within the duration of action of many sedation protocols, and faster image acquisition may allow scanning of the entire thorax within the sedated respiratory interval, thus making a combination of chemical and physical restraint a feasible option for patient immobilization and respiratory motion. As opposed to anesthesia, sedation has a lesser degree of respiratory muscle relaxation, potentially reducing compression atelectasis. Additionally, by allowing the patients to breathe room air as opposed to high FiO₂, diffusion atelectasis can be reduced. Recent investigations into the feasibility and diagnostic quality of CT scans performed on sedated or awake animals—upper airway studies in dogs, thoracic studies in cats and dogs, and abdominal studies in dogs—have demonstrated that diagnostic quality studies are achievable.^{21–25}

Objectives of the current study were to assess the image quality of thoracic CT scans performed with an MDCT scanner on dogs lacking pulmonary disease given injectable sedation by evaluating the lung volume, mean lung attenuation, and amount of poorly aerated and atelectatic lung. We hypothesized that scans performed using this method would have mean lung attenuation and lung volume measurements consistent with previously documented measurements for thoracic scans of anesthetized normal dogs.^{12,26,27} Additionally, we hypothesized that the amount of poorly aerated and atelectatic lung would be reduced compared to what has been documented for normal dogs under general anesthesia.¹¹

2 | MATERIAL AND METHODS

2.1 | Study design and subject inclusion

This was a prospective pilot study. Ten, client-owned dogs were sampled. All dogs referred to the Veterinary Imaging Center of San Diego for outpatient CT scans for non-thoracic pathology were recruited to participate during the period of May 1 to October 31, 2018. Informed consent was obtained from each owner. An ACVECC-certified veterinary critical care specialist evaluated each subject. Physical examinations, serum chemistries, and complete blood counts were performed and evaluated before the scans. Each subject's breed, age, sex, weight, and body condition score (BCS)²⁸ were recorded. Dogs were included if they had no clinical signs of respiratory disease at the time of the scan, had no history of respiratory disease, and were deemed healthy

enough for the sedation protocol. Dogs were excluded if there was thoracic pathology, such as pleural space or thoracic wall disease, and pulmonary masses or nodules noted on CT.

2.2 | Sedation protocol and computed tomographic image acquisition

An intravenous catheter was placed in either the cephalic or saphenous vein of each animal prior to the CT scans. The dogs were initially sedated with intravenous butorphanol (0.15–0.2 mg/kg) and diazepam (0.1–0.15 mg/kg). After five minutes, the dogs were given intravenous dexmedetomidine (3–6 µg/kg) for further immobilization. The subjects were kept in sternal recumbency while the medications took full effect. Once the subjects were acceptably immobilized, they were positioned in sternal recumbency on the CT bed. Positioning was aided by straps and foam rollers. Each patient was continuously visually monitored from the CT control station and directly auscultated in-between active scanning. After scan completion, each dog was given intramuscular atipamezole dosed at a volume equivalent to the total volume of dexmedetomidine.

The entire thorax of each animal was scanned three times at 8-minute intervals: time 0, time 8, and time 16. The images were obtained in a lung algorithm using tube voltage of 120 kVp, 150 mAs tube current, 2 mm slice thickness, 0.75 sec rotation time, and pitch of 1.09 by a 40-slice multidetector CT scanner (Philips Brilliance 40 channel CT scanner, Amsterdam, Netherlands). The first scan (time 0) was performed directly after proper positioning of the patient was obtained. The scans that were part of the subject's clinical investigation were performed between time 0 and time 8. The last scan (time 16) was initiated 16 min after the initiation of the first scan. Because each subject was undergoing a clinical investigation, each patient was administered intravenous nonionic organically bound iodinated contrast at a dose of 1.35 g iopamidol/kg; a maximum of 36.7 g iopamidol total dose was used for dogs weighing greater than 27 kg. The time between the dexmedetomidine administration and the first scan, the time between intravenous contrast administration and the second scan, and each thoracic scan's duration was recorded. The scan timing was chosen to allow for distribution and excretion of the contrast agent and allow the scans to be completed within the duration of action of the sedation protocol.

2.3 | Image analysis and segmentation

Subjective evaluations, measurements, and computer-generated segmentations for all scans were performed by a specialty intern with 1 year of diagnostic imaging experience (T.D.H.). Atelectasis on CT was based on previously published criteria and was defined as pulmonary regions with attenuation of –100 to 100 Hounsfield units (HU).¹⁵ To identify scans with regions of atelectasis, each thoracic scan was visually inspected for pulmonary regions of conspicuously increased attenuation. The images were reviewed in a lung window (WL: –500, WW:

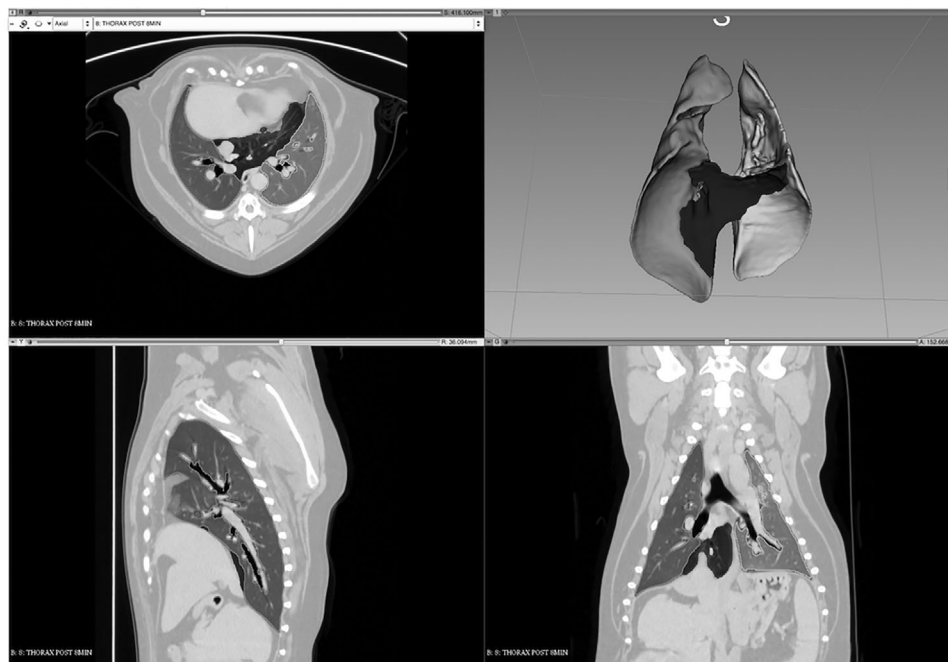


FIGURE 1 Example of the three-dimensional model and label map of the pulmonary parenchyma from which mean lung attenuation, total lung volume, and percentage of high attenuating lung area were generated

1400) using DICOM viewing software (Horos 3.3.5, Horos Project, Annapolis, MD USA) on a computer workstation (iMac 2017, Apple Inc., Cupertino, CA USA). The CT value of that region was measured with a circular region of interest (ROI) of 10–12 mm² using the DICOM viewing software's calipers. Pulmonary vasculature was excluded from the ROIs as much as possible.

Three-dimensional (3D) reconstruction software (3D Slicer Chest Imaging Platform, <http://www.slicer.org>) was used to obtain total lung volumes, mean lung attenuation, and percentage of high attenuating lung areas. Each scan was uploaded to the software, and masking using Hounsfield unit thresholds was used to identify anatomical structures and the boundaries between tissues. The threshold limits used were: 350 HU to –150 HU to identify the thoracic wall, soft tissue structures of the abdomen, the mediastinal structures, and pulmonary vasculature; –1025 HU to –900 HU to identify the trachea and primary through tertiary bronchi; –899 HU to –151 HU to identify the pulmonary parenchyma. Grow-from-seed processing was used to create a 3D model and label map of the entire thorax. The pulmonary parenchyma, exclusive of the airways and the vasculature, was isolated for analysis (Figure 1). The software calculated total lung volume and mean lung attenuation. Areas of pulmonary parenchyma with attenuation values greater than –500 HU and –250 HU were generated by the software to assess the regions of poorly aerated or hypoinflated lung, which has previously been defined CT values of –500 HU to –101 HU.¹⁸ These values were expressed as a percentage of total lung area for each subject.

2.4 | Statistical analysis

A specialty intern with a Master of Preventative Veterinary Medicine degree (T.D.H.) performed the following analyses, using commercially available software (Excel for Mac 16.45, Microsoft Corporation, Redmond, WA). Per kilogram lung volumes were calculated by dividing the total lung volume generated from the segmentation software by each patient's ideal body weight. Mean, standard deviation (SD), and range were generated for the variables of interest. Previously described data on mean lung attenuation, mean lung volume, and percent of high attenuation area were recorded. Two-sample *t*-test was used to compare current results to previously published data where mean and standard deviation was reported; one-sample *t*-test was used for data only reported as mean.

A statistical consultant with a Ph.D. in epidemiology performed the following analyses using statistics software (SAS 9.4, SAS Institute Inc., Cary, NC). Data were examined for normality by using the Shapiro-Wilk test. Variables that were not normally distributed were log-transformed before inclusion in the mixed linear regression. The association between the mean lung attenuation and the predictors (subject's body weight, lung volume, and time) were assessed by using adjusted mixed linear regression to account for the repeated measurements on individual animals. Mixed linear regression was also used to evaluate the association between other outcomes (lung volume and percentage of high attenuation lung area) and predictors (subject's body weight and time). *P*-values < 0.05 were considered significant.

TABLE 1 Results of unadjusted mixed linear regression analysis to assess the association of various predictors with total lung volume and per kg lung volume

Outcome	Predictor	Estimate	95% CI	P-value
Total lung volume (L)				
	<20 kg	Reference	–	–
	>30 kg	1.04	0.58, 1.51	0.0001*
	Weight (kg)	0.039	0.017, 0.061	0.0013*
	time 0	Reference	–	–
	time 8	–0.049	–0.13, 0.038	0.25
	time 16	–0.028	–0.11, 0.058	0.5
Per kg Lung volume (ml/kg)				
	<20 kg	Reference	–	–
	>30 kg	5.19	–13.01, 23.41	0.55
	Weight (kg)	0.13	–0.61, 0.86	0.72
	time 0	Reference	–	–
	time 8	–1.65	–5.43, 2.13	0.37
	time 16	0.44	–4.22, 3.34	0.8

Abbreviation: CI, confidence interval.

*Indicates statistical significance ($P < 0.05$).

3 | RESULTS

3.1 | Study population

Dog signalment data and clinical indications for CT examinations are listed in Supplement 1. The mean body weight $\pm$ SD of all the dogs was 23.7 ± 13.2 kg. Five dogs had a bodyweight of greater than 30 kg, and five weighed less than 20 kg. Four dogs had a normal BCS of 5 (on a 9-point scale); three dogs had BCS of 6 and 7, respectively. There was no difference with respect to BCS between the dogs that weighed > 30 kg and dogs that weighed < 20 kg.

For the seven dogs for which it was recorded, the time between the administration of the dexmedetomidine and the first scan was 16.7 ± 6.0 min (range 10.4–25.6 min). For all dogs, the average time between intravenous contrast administration and the second scan (time 8) was 5.5 ± 0.9 min (range 4.6–7.3 min). The average duration of the thoracic scan was 6.9 ± 1.5 s (range 4.8–9.7 s). All subjects recovered uneventfully, and no complications were experienced during or after the scans.

3.2 | Subjective assessment of atelectasis

No scans had regions of increased attenuation consistent with atelectasis (–100 to 100 HU). Seven of the 30 (23.4 %) scans performed had visibly increased attenuation areas of –300 HU to –200 HU. Ten (33.4%) had regions of increased attenuation of –400 HU to –300 HU. Nine (30%) had regions of increased attenuation from –500 HU

to –400 HU; and 4 (13.4%) had regions of increased attenuation from –600 HU to –500 HU.

3.3 | Lung volume and mean lung attenuation

The mean total lung volume $\pm$ SD for all dogs was 1.09 ± 0.63 L. The mean total lung volume of dogs weighing > 30 kg (1.61 ± 0.31 L) was significantly greater ($P = 0.0001$) than the mean total lung volume of dogs weighing < 20 kg (0.56 ± 0.36 L). The mean per kg lung volume $\pm$ SD or all dogs was 50.9 ± 13.2 ml/kg. The per kg lung volume was not significantly different ($P = 0.55$) between dogs weighing > 30 kg (53.5 ± 10.4 ml/kg) and dogs weighing < 20 kg (48.3 ± 15.5 ml/kg). The total lung volume and per kg lung volume did not significantly change over the three time points (Table 1).

The mean lung attenuation $\pm$ SD at time 0, time 8, and time 16 was -707.0 ± 60.0 HU, -692.2 ± 56.4 HU, and -695.4 ± 58.4 HU, respectively. For dogs weighing > 30 kg, the mean lung attenuation $\pm$ SD at time 0, time 8 and time 16 was -743.0 ± 44.3 HU, -723.8 ± 44.8 HU, and -725.3 ± 43.5 HU, respectively. For dogs weighing < 20 kg, the mean lung attenuation $\pm$ SD at time 0, time 8 and time 16 was -671.0 ± 53.8 HU, -660.7 ± 51.5 HU, and -665.5 ± 59.5 HU, respectively (Figure 2).

Mean lung attenuation was significantly higher at time 8 and time 16 compared to time 0, but not statistically different between time 8 and time 16. Mean lung attenuation was significantly negatively correlated with total lung volume and per kg lung volume, yet the association between mean lung attenuation and weight, either as a

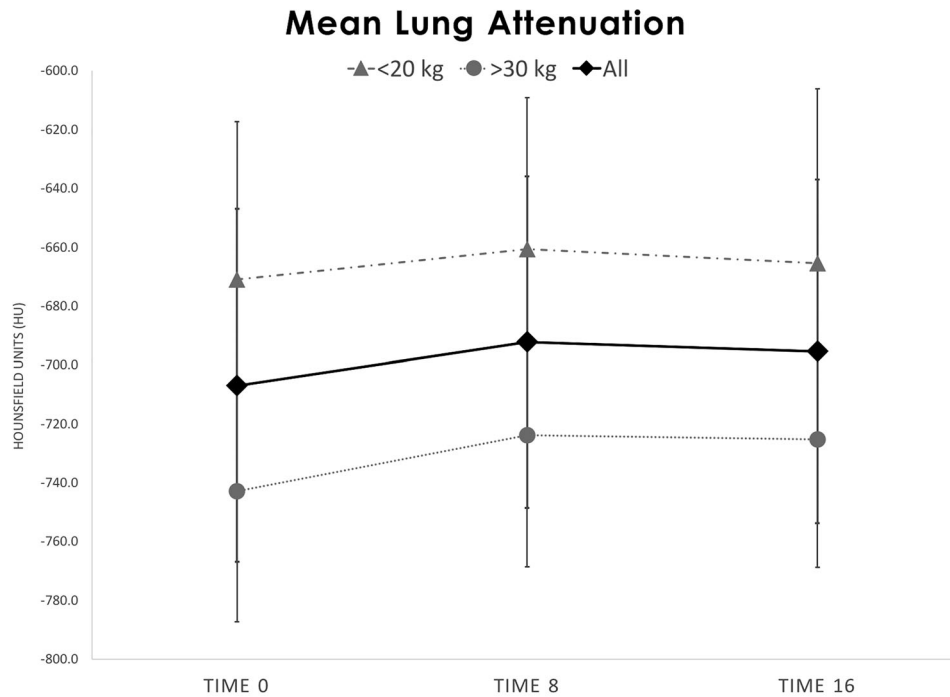


FIGURE 2 Line graph of the mean lung attenuation $\pm$ SD off all subjects ($n = 10$), dogs weighing < 20 kg ($n = 5$) and dogs weighing > 30 kg ($n = 5$) measured at three time points. The mean lung attenuation at time 8 and time 16 was statistically different compared to time 0; but not significantly different between time 8 and time 16. There was no statistical difference between the dogs weighing < 20 kg and dogs weighing > 30 kg

TABLE 2 Results of adjusted mixed linear regression analysis to assess the association of various predictors with mean lung attenuation

Outcome	Predictor	Estimate	95% CI	P-value
Mean lung attenuation (HU)				
	<20 kg	reference	–	–
	>30 kg	–64.94	–129.7, –0.15	0.05
	weight (kg)	–2.41	–5.08, 0.24	0.07
	time 0	reference	–	–
	time 8	11.25	2.76, 19.73	0.012*
	time 16	9.56	1.16, 17.97	0.028*
	time 16	reference	–	–
	time 8	1.68	–6.7, 10.1	0.67
	Total lung volume (ml)	–71.74	–100.73, –42.74	<.0001*
	per kg lung volume (ml/kg)	–2.06	–3.22, –0.90	0.0015*

Abbreviation: CI, confidence interval; HU, Hounsfield unit.

*Indicates statistical significance ($P < 0.05$).

categorical variable or continuous variable, was not statistically significant (Table 2).

3.4 | Percent of poorly aerated lung

The mean $\pm$ SD percent of high attenuating lung area greater than -500 HU (HAA%–500) at time 0, time 8 and time 16 was $10.7 \pm$

4.7%, $11.4 \pm 4.4\%$, and $11.7 \pm 5.1\%$, respectively. For dogs weighing > 30 kg, the mean $\pm$ SD HAA%–500 at time 0, time 8 and time 16 was $8.1 \pm 2.1\%$, $8.5 \pm 2.1\%$, and $9.6 \pm 3.0\%$, respectively. For dogs weighing < 20 kg, the mean $\pm$ SD HAA%–500 at time 0, time 8 and time 16 was $13.3 \pm 5.4\%$, $14.4 \pm 4.2\%$, and $13.9 \pm 6.1\%$, respectively (Figure 3A).

The mean $\pm$ SD percent of lung area with attenuation greater than -250 HU (HAA%–250) at time 0, time 8, and time 16 was $2.4 \pm 1.2\%$,

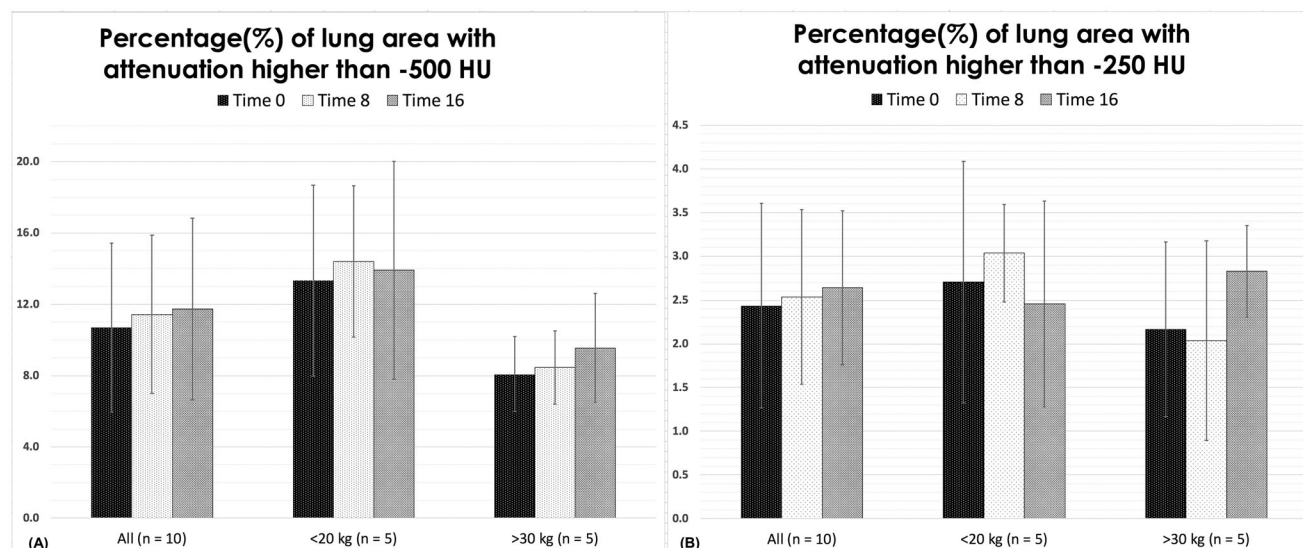


FIGURE 3 Bar graph of the mean $\pm$ SD percentage of lung area with attenuation values A, greater than -500 HU and B, greater than -250 HU for all subjects and for subjects grouped by body weight over the three time points. The percentage of high attenuating lung had no statistically significant association with time or body weight. Abbreviations: HU, Hounsfield unit

TABLE 3 Results of unadjusted mixed linear regression analysis to assess the association of various predictors with the respective percentage of high attenuating lung area

Outcome	Predictor	Estimate	95% CI	P-value
HAA%–500				
	<20 kg	Reference	–	–
	>30 kg	–5.17	–10.38, 0.04	0.052
	Weight (kg)	–0.19	–0.41, 0.016	0.068
	time 0	Reference	–	–
	Time 8	0.73	–0.58, 2.05	0.25
	time 16	1.03	–0.27, 2.35	0.11
HAA%–250				
	<20 kg	Reference	–	–
	>30 kg	–0.39	–1.51, 0.73	0.47
	Weight (kg)	–0.013	–0.058, 0.032	0.55
	time 0	Reference	–	–
	Time 8	0.1	–0.58, 0.79	0.76
	time 16	0.2	–0.48, 0.89	0.53

Note: P-value of < 0.05 is considered significant. Abbreviation: CI, confidence interval; HAA%–500, percentage of lung area with attenuation greater than -500 Hounsfield units; HAA%–250, percentage of lung area with attenuation greater than -250 Hounsfield units.

$2.5 \pm 1.0\%$, and $2.6 \pm 0.9\%$, respectively. For dogs weighing > 30 kg, the mean $\pm$ SD HAA%–250 at time 0, time 8, and time 16 was $2.2 \pm 1.0\%$, $2.0 \pm 1.1\%$, and $2.8 \pm 0.5\%$, respectively. For dogs weighing < 20 kg, the mean $\pm$ SD HAA%–250 at time 0, time 8, and time 16 was $2.7 \pm 1.3\%$, $3.0 \pm 0.5\%$, and $2.5 \pm 1.2\%$, respectively (Figure 3B). There was no significant change in HAA%–500 nor HAA%–250 over the three time points and no association between weight and high attenuation area (Table 3).

3.5 | Comparison to prior published data

In a previous study,¹¹ six groups of 10 dogs underwent thoracic CT performed under general anesthesia and mechanical ventilation. When our results for hypoinflated lung (HAA%–500 and HAA%–250) are compared to the prior study's values, the current study's values are statistically lower (Table 4). The group of dogs from the prior study given 15 cmH₂O positive pressure ventilation (PP) and the current

TABLE 4 Two-sample *t*-test *P*-values comparing mean $\pm$ standard deviation percentage of high attenuating lung area of the current study to areas of hypoinflation and atelectasis from previously published data. Significant *P*-values indicate difference between variables

Outcome	Prior study ^a 0 cm H ₂ O PP	Prior study ^a 15 cm H ₂ O PP
HAA%–500	19.8 $\pm$ 5.0% (n = 10)	11.1 $\pm$ 1.9% (n = 10)
Total (n = 10) 10.7 $\pm$ 4.7%	0.0005 [*]	0.81
>30 kg (n = 5) 8.1 $\pm$ 2.1%	2.5 $\times$ 10 ^{–5} [*]	0.019 [*]
<20 kg (n = 5) 13.3 $\pm$ 5.4%	0.042 [*]	0.39
HAA%–250	5.0 $\pm$ 2.3% (n = 10) ^b	2.8 $\pm$ 0.9% (n = 10) ^b
Total (n = 10) 2.4 $\pm$ 1.2%	0.005 [*]	0.41
>30 kg (n = 5) 2.2 $\pm$ 1.0%	0.006 [*]	0.27
<20 kg (n = 5) 2.7 $\pm$ 1.3%	0.028 [*]	0.88

Note:

^{*}Indicates statistical significance (*P* < 0.05).

^aGuarracino et al.¹¹

^bValues represent percentage of lung area with attenuation greater than –100 Hounsfield units (HU) and less than 100 HU; HAA%–500, percentage of lung area with attenuation greater than –500 HU; HAA%–250, percentage of lung area with attenuation greater than –250 HU; (n), number of dogs per group; PP, constant ventilation positive pressure.

TABLE 5 Two-sample *t*-test *P*-values comparing mean $\pm$ standard deviation lung attenuation compared to previously published data. Significant *P*-values indicate difference between variables

Outcome	Prior study (n = 46) ^a –774.0 $\pm$ 43.5 HU	Prior study (n = 28) ^b –775 $\pm$ 30 HU	Prior study (n = 28) ^b –856 $\pm$ 22 HU
Total (n = 10) 707.0 $\pm$ 60.0 HU	0.0015 [*]	0.0015 [*]	4.4 $\times$ 10 ^{–9} [*]
>30 kg (n = 5) 743.0 $\pm$ 44.3 HU	0.14	0.13	4.1 $\times$ 10 ^{–6} [*]
<20 kg (n = 5) 671.0 $\pm$ 53.8 HU	0.0001 [*]	0.0002 [*]	1.5 $\times$ 10 ^{–8} [*]

Abbreviations: HU, Hounsfield units; (n), number of dogs per group.

^{*}Indicates statistical significance (*P* < 0.05).

^aChoi et al.²⁵

^bAllison et al.¹²

study's population had statistically similar areas of hypoinflation, with the exception of the subgroup of dogs in the current study weighing > 30 kg, which had statistically lower HAA%–500.

Mean lung attenuation of normal anesthetized dogs has been reported in previous studies as –846 HU,²⁶ –774.0 $\pm$ 43.5 HU,²⁷ –775 $\pm$ 30 HU and –856 $\pm$ 22 HU¹² using different measurement techniques and variable positive pressure ventilation. The mean lung attenuation of the current study's population was significantly higher than the most recently published report's values.^{12,27} Except, the subgroup of dogs weighing > 30 kg did not have mean lung attenuation significantly different than the previous higher mean lung attenuation values (Table 5). Expiratory mean lung attenuation has previously been reported as –713 HU.³² No statistical difference was found between the mean lung attenuation for all dogs, dogs weighing > 30 kg and those weighing < 20 kg and the previously reported expiratory lung attenuation value (*P* = 0.76, *P* = 0.20, and *P* = 0.16, respectively).

Due to differences in study populations from previous studies, a comparison of total lung volume could not be made. The current study's per kg lung volume (50.9 $\pm$ 13.2 ml/kg) is not statistically different (*P* = 0.83) from a prior study's results (52 $\pm$ 14 ml/kg).¹²

4 | DISCUSSION

This study showed that the amount of hypoinflated lung identified on thoracic CT of sedated dogs was less extensive than what has been reported to be present in anesthetized dogs given no positive pressure ventilation and consistent with dogs given 15 cm H₂O PP.¹¹ Areas of severely hypoinflated lung (HAA%–250) were less extensive than the amount of atelectasis reported in anesthetized dogs given no PP and consistent with the amount of atelectasis present in dogs given 15 cm H₂O PP.¹¹ Regions of atelectasis were not subjectively identified. We demonstrated per kilogram lung volume was consistent with values previously reported for anesthetized dogs given no end-expiratory pressure.¹² Therefore, findings supported our hypothesis that CT scans of the canine thorax performed with a MDCT scanner under injectable sedation have less hypoinflation and no difference in lung volume than what has previously been described for normal anesthetized dogs. Additionally, consistent with prior studies,^{29–31} the current study showed that hypoinflated lung areas did not progress over time. Findings, therefore, supported using sedation for patient immobilization and positioning during CT scanning.

The previous studies that reported mean lung attenuation utilized various methods of measurement and employed various ventilatory pressures. In the oldest study,²⁶ mean lung attenuation was averaged from a sampling of small ROIs, while the latter two studies^{12,27} used threshold limited grow-from-seed volume reconstruction. Two previous studies^{26,27} performed CT scans at full inspiration while either applying 15 cm H₂O PP²⁶ or 10–15 cm H₂O PP.²⁷ The more recent study,¹² employed hyperventilation to induce apnea, variable positive end-expiratory pressure (PEEP), and intermittent 8–12 cm H₂O PP above PEEP. No published description of methods for acquiring the attenuation value for expiratory lung³² could be found. The different techniques and ventilatory protocols of these studies make comparisons between studies difficult, but it is apparent that mean lung attenuation is affected by insufflation and phase of respiration. Our results for mean lung attenuation were higher than these previous reports,^{12,26,27} but not statistically different from the reported expiratory lung values. The higher values of mean lung attenuation reported in the current study reflect an average expiratory timing of the scans.

Interestingly, the group of dogs weighing > 30 kg had statistically similar attenuation values to dogs given lower ventilatory pressures^{12,27} and had lower mean lung attenuation and lower percentages of poorly aerated lung than those weighing < 20 kg. However, we did not find a statistical difference between the dogs grouped by weight with respect to mean lung attenuation and percentages of poorly aerated lung. A previous study²⁷ demonstrated a weak but significant, negative correlation between weight and mean lung attenuation. The authors postulated the relatively lower amounts of lung tissue per unit volume of larger lungs and the positive correlation between lung compliance and bodyweight likely explains this correlation.

Although the current study demonstrated that the percent of high attenuating areas did not progress, there was a significant increase in mean lung attenuation over the 16 minutes study interval. We also noted that total lung volume did not progress over time. Because there was a significant association between total lung volume and mean lung area, we propose that the increase in mean lung attenuation over time was due to the administration of contrast agent prior to the second and third scans rather than a reduction of lung aeration.

The thresholds chosen to create the tissue mask identified the boundary between the soft tissue of the mediastinum and the pulmonary parenchyma. There is overlap between the attenuation of atelectatic lung and soft tissue, so some atelectatic lung was potentially excluded from the parenchymal analysis by using these thresholds. However, the grow-from-seed function is not constrained by the thresholds of the mask, only the boundaries delineated by it. Therefore, high attenuation voxels would be included in the pulmonary parenchyma map and included in the software-generated analysis. Additionally, we subjectively evaluated areas of conspicuously increased attenuation and measured high attenuation areas with ROIs.

This study had several limitations. The study design did not include a group of dogs scanned under general anesthesia, but instead relied on previously published data for comparisons. The differences in study protocols and results reporting make comparison inexact. Using a case-control match or performing scans separately under sedation and anes-

thesia on the same subject should be considered for future studies. Our sample size was small, using client-owned animals presenting for other clinical concerns, and the subject's body weights were not normally distributed. There was a lack of significance between parameters such as body weight and mean lung attenuation, and percentage of poorly aerated, despite a notable difference. A future area of study would be to evaluate a larger sample of dogs grouped by multiple weight categories to evaluate for differences among these populations. Lastly, the population was biased to geriatric dogs with no clinical evidence of lung disease. It was beyond the scope of this study to cytologically or histologically confirm that the study population of dogs lacked pulmonary pathology.

The current study focused on the evaluation of normal lung and expected values of attenuation. This study found that sedation performed satisfactorily without complication for thoracic CT of dogs without pulmonary disease. Diseased lung may display different characteristics than what has been described here. In a prior study, comparing thoracic CT findings performed under sedation versus anesthesia on West Highland White Terriers with idiopathic pulmonary fibrosis, differences in the identification of ground-glass opacification and mosaic attenuation pattern were recorded in less than 50% of affected dogs.³³

Prior studies reported a gradation system for motion artifact on thoracic CT. In one study, motion was ranked as barely noticeable to mild in all dogs and was not statistically different between normal dogs and dogs with diseased lungs when both groups were scanned awake or with sedation.²⁵ In another study, a statistical difference between motion artifact present on diseased dogs scanned under sedation versus those scanned under general anesthesia was found.³³ Yet in both groups, the majority of the scans were graded as mild. These prior studies recorded grading schemes that partially incorporated an assessment of interpretability, limiting their use in the current study. In future studies, a more objective grading system could be devised and analysis of variables such as motion and alteration of attenuation should be assessed.

Additionally, an important limitation to sedated thoracic CT is the inability to time the scans with inspiration and expiration. Indicators of pulmonary disease such as air trapping, dynamic airway collapse, and bronchiectasis may be more conspicuous on paired inspiratory/expiratory scans and expiratory scan may underestimate the severity of these tomographic findings.³⁴ Yet, recent studies demonstrated thoracic CT on sedated dogs was superior to radiographs at detecting evidence of blunt force trauma after vehicular accidents²⁴ and there is good agreement between abnormal CT findings and subject's clinical status in awake White Highland White Terriers with idiopathic pulmonary fibrosis.²⁵ Future studies should also be aimed at evaluating the indications for sedated CT scans and different sedation protocols to accommodate patients with medication limitations based on comorbidities.

In conclusion, thoracic CT scans performed under sedation in this sample of dogs displayed comparable lung characteristics and qualities described for CT performed on normal dogs under general anesthesia. Based on this study, we believe sedation is a sufficient method of

restraint when performing thoracic CT in dogs, does not itself lead to alterations of lung attenuation, and may confer less hypoinflation and atelectasis than general anesthesia.

ACKNOWLEDGMENTS

The authors acknowledge Christina A. Hewes, Veterinary Imaging Center of San Diego, for clinical support and case management; and Chunyi Wu, Institute of Gerontology, University of Michigan Medical School, Ann Arbor, MI for statistical consultation and computation.

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Category 1

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- (b) Acquisition of Data: Hunt
- (c) Analysis and Interpretation of Data: Hunt

Category 2

- (a) Drafting the Article: Hunt
- (b) Revising Article for Intellectual Content: Hunt, Wallack

Category 3

- (a) Final Approval of the Completed Article: Hunt, Wallack

CONFLICT OF INTEREST

The authors have declared no conflict of interest.

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SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Hunt TD, Wallack ST. Minimal atelectasis and poorly aerated lung on thoracic CT images of normal dogs acquired under sedation. *Vet Radiol Ultrasound*. 2021;62:647-656. <https://doi.org/10.1111/vru.13016>